CHEMISTRY
A EUROPEAN JOURNAL

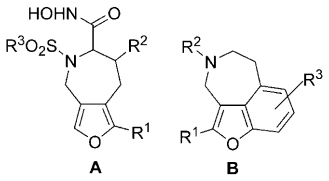
DOI: 10.1002/chem.201003546

Selectivity Manipulation of Bicyclization Reactions of 1,5-Bis(1,2-allenylketone)s: Pd versus Rh and Electronic Effect

Youqian Deng, Chunling Fu, and Shengming Ma*[a]

In memory of S. Ma's father, Mr. Xinwei Ma, who passed away last September

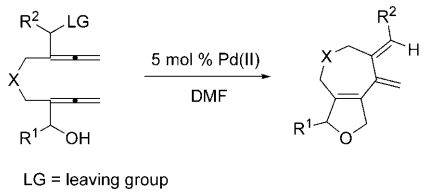
Recently, much attention has been paid to the synthesis of furan-fused bicyclic compounds[1–4] due to their importance in medicinal chemistry. For example, it has been reported that the furan-fused cycloheptanes, such as TNF-α inhibitors (Scheme 1A)[5] and α-adrenergic receptor antagonists (Scheme 1B),[6] are a class of compounds with important pharmacological activities. It should be noted that the presence of the nitrogen atom in the seven-membered ring is crucial for the biological activity of this type of compounds. Thus, it is highly desirable to develop efficient methods that may include different carbon or heteroatom units into the seven-membered ring for further biological screening.

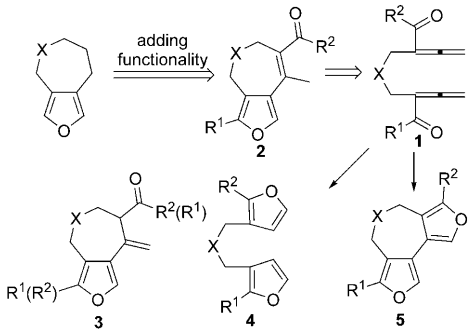


Scheme 1. Furan-fused bicyclic or tricyclic compounds with pharmacological activities.

On the other hand, recently, we have developed a palladium(II)-catalyzed tandem bicyclization reaction of symmetrical 1,ω-bisallenols to form 2,5-dihydrofuran-fused bicyclic skeletons, in which the second OH moiety in the second allene unit serves as the leaving group after intramolecular oxypalladation-carbopalladation to finally form the two *exo*-cyclic C=C bonds [Eq. (1)].[7]

The transition-metal-catalyzed monocyclization of simple 1,2-allenyl ketones has been well studied by Marshall, Hashmi, Gevorgyan, and our group.[8,9] With this in mind, we reasoned that furan-fused cycloheptenes would be prepared efficiently from a Pd^{II}-catalyzed cyclization reaction of 1,5-bis(1,2-allenylketone)s, providing that the second allenyl ketone unit would serve as the Michael acceptor, instead of the β-heteroatom (LG) elimination in Equation (1).[10] If this is viable, the carbon or heteroatom unit, which used as the tether to link the two allenyl ketone units, would be easily assembled into the seven-membered ring (Scheme 2). However, based on Hashmi's early observation,[10] we envisioned that there are three major challenges of this strategy: 1) the issue of selectivity for the location of the nonaromatic C=C bond in the products (forming products **2** and/or **3**); 2) the control of selectivity for the unsymmetrical substrates (R^1 versus R^2 in products **2** and **3**); and in addition, 3) other possible products would be dumb-bell-type bisfurans **4**[8a,9,10] and tricyclic products **5** shown in Scheme 2. Herein we wish to report a rhodium(I)-catalyzed double cyclization of 1,5-bis(1,2-allenylketone)s that afforded diversified furan-fused bicyclic products **2** with a very high selectivity.





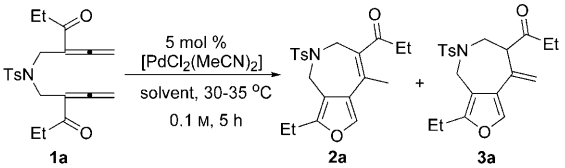
Scheme 2. Possible products of the cyclization of bis(allenylketone)s **1**, illustrating the selectivity problems (X = carbon or heteroatom).

[a] Dr. Y. Deng, Prof. Dr. C. Fu, Prof. Dr. S. Ma
Laboratory of Molecular Recognition and Synthesis
Department of Chemistry, Zhejiang University
Hangzhou 310027, Zhejiang (P. R. China)
Fax: (+86) 21-64167510
E-mail: masm@sioc.ac.cn

Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/chem.201003546.

WILEY ONLINE LIBRARY
© 2011 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Table 1. Solvent effects on the Pd^{II}-catalyzed cyclization reaction of 1,5-bis(1,2-allenylketone)s **1a** (Ts = tosyl).

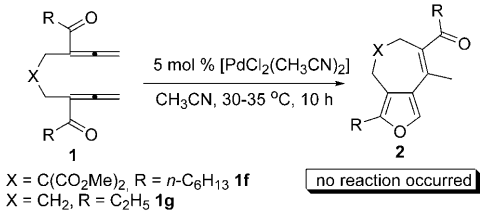


	Solvent	NMR yield of (**2a**+**3a**) [%][a]
1	DMF	16 (69/31)[b]
2	DMA	21 (89/19)[b]
3	DMSO	53 (83/17)[b]
4	CH_3NO_2	trace
5	THF	26 (85/15)[b]
6	CH_3CN	68 (90:10)[b]
7	acetone	54 (83:17)[b,c]

[a] Determined by ^1H NMR analysis with 1,3,5-trimethylbenzene as the internal standard. [b] NMR ratio of **2a**/**3a**. [c] The reaction was conducted at RT.

Initially, we tested the reaction of *N*-tethered 1,5-bis(1,2-allenylketone)s **1a** catalyzed by 5 mol % of $[PdCl_2(MeCN)_2]$ in different solvents (Table 1).[7] Although the results in DMF, dimethylamide (DMA), DMSO, CH_3NO_2, and THF are poor (Table 1, entries 1–5), the reaction in CH_3CN proceeded smoothly to afford furo[3,4-*c*]azepine derivative **2a** together with the C=C bond regioisomer **3a** with a ratio of 90:10 in 68% combined yield (Table 1, entry 6).

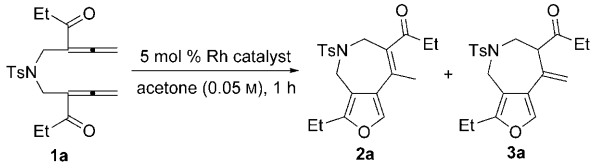
To improve the selectivity of **2a**/**3a**, we tried many reaction parameters; however, the results were not improved. It is even difficult to convert **3a** into **2a,** by adding a base or an acid, without a big loss in yield. In addition, the scope of this Pd-catalyzed reaction is extremely limited, because no reaction occurred with **1f** and **1g**, which had different carbon tethers (Scheme 3).



Scheme 3. The very limited scope of the Pd^{II}-catalyzed protocol.

Thus, we started to check the possible use of Rh^I catalysts to address the issue of regioselectivity (**2a** versus **3a**) and the scope of the reaction. Disappointingly, $[RhCl(PPh_3)_3]$[8a] and $[RhCl(CO)(PPh_3)_2]$ failed to afford the expected product **2a** in acetone (Table 2, entries 1 and 2). But luckily and as expected, when $[RhCl(COD)_2]_2$ was used, **2a** was formed in 52% NMR yield along with **3a** in a higher selectivity of 95:5 (**2a**/**3a**, Table 2, entry 3). Furthermore, the selectivity was improved to >99:1 when the reaction was conducted in acetone under reflux with $[RhCl(CO)_2]_2$ as the catalyst to give a yield of 85% (Table 2, entry 4). Finally, screening the

Table 2. Catalyst effects on the Rh^I-catalyzed cyclization reaction of 1,5-bis(1,2-allenylketone)s **1a**.



	Catalyst (5 mol %)	*T*	**2a**/**3a**	NMR yield of **2a** [%]
1	$[RhCl(PPh_3)_3]$	reflux	–	0[a]
2	$[RhCl(CO)(PPh_3)_2]$	reflux	–	0[b]
3	$[RhCl(COD)]_2$	reflux	95:5	52
4	$[RhCl(CO)_2]_2$	reflux	>99:1	85
5	$[RhCl(CO)_2]_2$	RT	>99:1	83 (80)[c]

[a] 95% of **1a** was recovered. [b] 71% of **1a** was recovered. [c] Isolated yield.

reaction temperature showed that the reaction may proceed smoothly even at RT, which was defined as the standard conditions (Table 2, entry 5). The double-cycloisomerization product **4a** and triple-cyclization product **5a** (Scheme 2) were not observed even under the catalysis of $[RhCl(PPh_3)_3]$ (Table 2, entry 1).[8a] The structure of **2a** was further established by X-ray diffraction studies (Figure 1).[11] Here, a liable ligand is clearly required to allow the cyclization, which should involve the coordination of the C=C bonds with the metal. As a comparison, under this set of new conditions with $[PdCl_2(MeCN)_2]$ as the catalyst, the reaction still afforded a mixture (Table 1, entry 7).

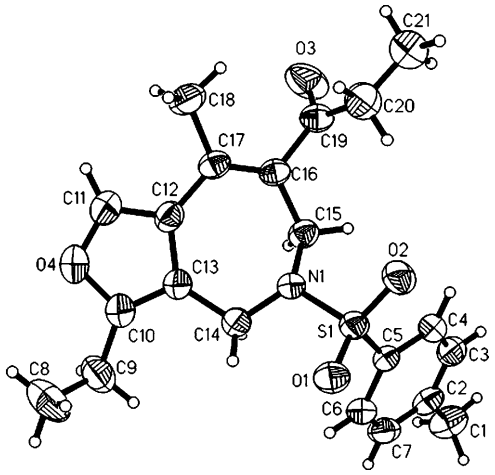


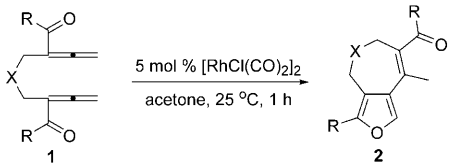
Figure 1. ORTEP representation of **2a**.

With this set of optimized reaction conditions in hand, the scope of the cyclization reactions was demonstrated and some typical results are summarized in Table 3. The reaction of 1,5-bis(1,2-allenylketone)s **1a**, **1b**, **1c**, and **1d** with NTs as the tether yielded furo[3,4-*c*]azepines **2a**, **2b**, **2c**, and **2d** in good yields (Table 3, entries 1–4). The sulfone tether may also be introduced (Table 3, entry 5). Fortunately, the 1,5-bis(1,2-allenylketone)s **1f** and **1g**, with a carbon tether,

 © 2011 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

15213765, 2011, 18, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201003546 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Substrate scope for the cyclization reaction of 1,5-bis(1,2-allenylketone)s **1**.



	X	R	Isolated yield of **2** [%]
1	NTs	C_2H_5 (**1a**)	80 (**2a**)
2	NTs	CH_3 (**1b**)	84 (**2b**)
3	NTs	*i*-C_3H_7 (**1c**)	77 (**2c**)
4	NTs	*n*-C_5H_{11} (**1d**)	75 (**2d**)
5	SO_2	C_2H_5 (**1e**)	76 (**2e**)
6[a]	$C(CO_2Me)_2$	*n*-C_6H_{13} (**1f**)	77 (**2f**)
7	CH_2	C_2H_5 (**1g**)	65 (**2g**)
8[b]	CH_2	*i*-C_3H_7 (**1h**)	72 (**2h**)
9	CH_2	*n*-C_4H_9 (**1i**)	66 (**2i**)
10[c]	CH_2	Ph (**1j**)	44 (**2j**)
11[c]	CH_2	Ph (**1j**)	66 (**2j**)

[a] The reaction was conducted under reflux. [b] The reaction time is 2 h. [c] The reaction was conducted in 1,2-dichloroethane (DCE) at 70 °C for 1 h.

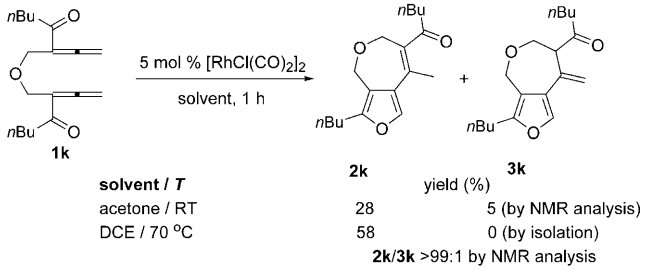
fails to proceed under the catalysis of $[PdCl_2(MeCN)_2]$ (Scheme 2), but react smoothly under the catalysis of $[RhCl(CO)_2]_2$ (Table 3, entries 6 and 7), although a higher reaction temperature was required when 1,5-bis(1,2-allenylketone) **1f**, with a malonate tether, was used (Table 3, entry 6). The seven-membered carbon ring seems to be preferred, since 1,5-bis(1,2-allenylketone)s **1h** and **1i** with a CH_2 tether under the standard conditions provide the fused [5.3.0]bicyclic products **2h** and **2i** in good yields (Table 3, entries 8 and 9). By applying this protocol, the reaction of phenyl-substituted 1,5-bis(1,2-allenylketone) **1j** afforded product **2j** in 44% yield (Table 3, entry 10). ^{1}H NMR spectra of the crude products indicated the formation of only one product.

However, the reaction of 1,5-bis(1,2-allenylketone)s **1k**, with the oxygen tether, failed to afford the expected product **2k** effectively and selectively under the standard conditions. Further screening led to the observation that the fused [5.3.0]bicyclic product **2k** was formed smoothly in 58% isolated yield when the reaction was conducted in DCE at 70 °C (Scheme 4). With this new set of reaction conditions, the yield of **2j** was also improved (Table 3, entry 11).

Furthermore, unsymmetrical diketones selectively afford products **2l** and **2m** with *n*-butyl or benzyl allenyl ketone cyclization and the phenyl or *p*-nitrophenyl allenyl ketone serving as the Michael acceptor (Table 4). A combination of *n*Bu or *i*Pr with *p*-nitrophenyl even led to an exclusive formation of **2n** and **2o**. The selectivity observed here has been established by X-ray diffraction studies of **2o** (Figure 2).[12]

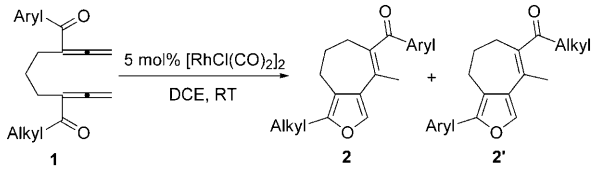
Surprisingly, even for **1p** with a *p*-nitro and *p*MeO group on each phenyl ring, respectively, the electron-donating aryl allenyl ketone cyclizes, whereas the electron-withdrawing aryl allenyl ketone accepts the conjugate addition from the in situ formed furanyl–rhodium intermediate to selectively form of products **2p** and **2p′** in 55% and 7% isolated yields, respectively (Scheme 5).

To further unveil the nature of the selectivity observed with the rhodium catalyst, a mixture of **2a** and **3a** in a ratio



Scheme 4. The reaction of the oxygen-tethered **1k**.

Table 4. Reactions of aromatic–aliphatic-mixed 1,5-bis(1,2-allenyl ketone)s **1**.



	Aryl	Alkyl	*t* [min]	Yield of **2** [%][a]	Yield of **2′** [%][a]
1	Ph	*n*Bu (**1l**)	20	74 (**2l**)	8 (**2l′**)
2	*p*-$O_2NC_6H_4$	Bn[c] (**1m**)	60	55 (**2m**)	3[b] (**2m′**)
3	*p*-$O_2NC_6H_4$	*n*Bu (**1n**)	20	77 (**2n**)	0 (**2n′**)
4	*p*-$O_2NC_6H_4$	*i*Pr (**1o**)	20	60 (**2o**)	0 (**2o′**)

[a] Isolated yield. [b] NMR yield. [c] Bn = benzyl.

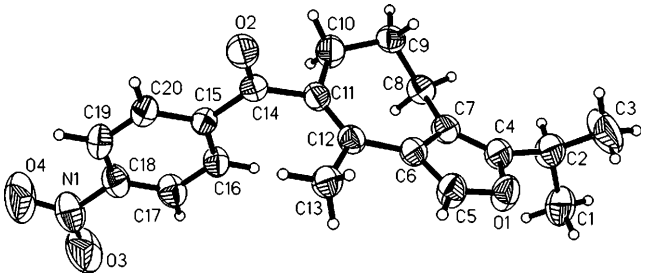
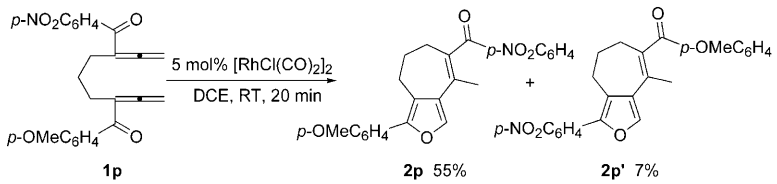


Figure 2. ORTEP representation of **2o**.

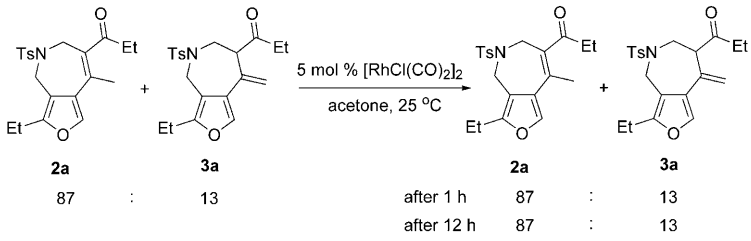


Scheme 5. Reactions of the unsymmetrical 1,5-bis(1,2-allenylketone)s **1p**.

© 2011 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

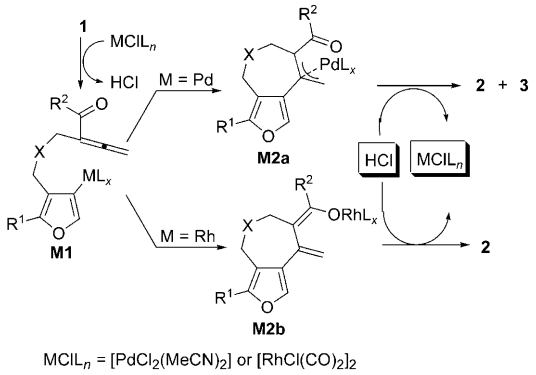
15213765, 2011, 18, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201003546 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of 87:13 was treated with 5 mol% $[RhCl(CO)_2]_2$ for 1 h under the standard reaction conditions or even 12 h with an unchanged **2a**/**3a** ratio (Scheme 6). This clearly indicates



Scheme 6. Possible isomerization between **2a** and **3a** under the Rh^I catalysis.

that the excellent selectivity observed with rhodium was not caused by isomerization between α,β-unsaturated enones **2** and β,γ-unsaturated enones **3**; it is probably owed to the direct protonolysis of the oxygen-bound rhodium dienolate intermediate shown in Scheme 7.



Scheme 7. Possible mechanism for the formation of **2** and **3**.

The formation of the 3,4-fused bicyclic furan skeletons may be rationalized by the mechanism shown in Scheme 7. Firstly, the cyclic oxymetallation of the 1,2-allenyl ketone moiety in **1** with Pd^{II} or Rh^I would form intermediate **M1** (M = Pd or Rh). The electron-donating R^1 group increases the nucleophilicity of the carbonyl oxygen atom; this explains the selectivity for unsymmetrical substrates with Rh^I catalysis. Then intramolecular carbometallation of the remaining allene unit in **M1** would form the π-allylic palladium intermediate **M2a** due to low oxophilicity of palladium.[13] Due to the presence of the carbonyl group, subsequent protonolysis of **M2a** with H^+ may occur. However, such a reaction at the α or γ-position would afford an isomeric mixture of **2** and **3**, with **2** as the major product due to the steric effect, and regenerate the catalytically active Pd^{II} species. However, it has been reported that rhodium prefers oxygen bonding in the enolate-type intermediates.[14] Thus, we reasoned that by using the rhodium catalyst oxygen-bound rhodium dienolate intermediate **M2b** must have formed exclusively; this yields **2** as the only product.

In conclusion, we have developed an efficient and highly selective synthesis for 3,4-fused bicyclic furans through a rhodium(I)-catalyzed cycloisomerization of readily available 1,5-bis(1,2-allenylketone)s[15] under mild conditions. Compared with $[PdCl_2(MeCN)_2]$, $[RhCl(CO)_2]_2$ shows an excellent selectivity at RT; this may be explained by the different bonding nature of palladium and rhodium to the carbon or oxygen in the organometallic intermediates. Because reports on such observations are very limited,[13,14] this will also be very informative for the future design of the reactions involving Pd and Rh enolates or dienolates. In addition, the selectivity of unsymmetric substrates has been nicely controlled by the electronic effect of nonallenyl groups connected to the carbonyl functionalities. Due to the potentials of the products and the excellent selectivities observed, this method will also be of high interest to organic and medicinal chemists. Further studies concerning the mechanism and synthetic applications in this area are being pursued in our laboratory.

Experimental Section

$[RhCl(CO)_2]_2$-catalyzed cyclization of 1: Acetone (2 mL) was added to a mixture of $[RhCl(CO)_2]_2$ (2.2 mg, 0.0057 mmol) and **1a** (38.0 mg, 0.10 mmol) at RT. The reaction was stirred at RT for 1 h. After the reaction had run to completion, as monitored by TLC, evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate 5:1) afforded **2a** (30.3 mg, 80%) as a solid, m.p.: 110–111 °C (ethyl acetate/hexane); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.62$ (d, $J = 8.1$ Hz, 2H), 7.27 (s, 1H), 7.22 (d, $J = 8.1$ Hz, 2H), 4.31 (s, 2H), 4.04 (s, 2H), 2.68 (q, $J = 7.2$ Hz, 2H), 2.48 (q, $J = 7.2$ Hz, 2H), 2.39 (s, 3H), 2.03 (s, 3H), 1.11 ppm (t, $J = 7.2$ Hz, 6H); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 205.6$, 153.4, 143.3, 139.5, 137.3, 135.8, 131.2, 129.4, 127.6, 127.2, 112.5, 47.2, 42.7, 35.4, 21.4, 19.7, 19.4, 12.4, 8.0 ppm; IR (neat): $\tilde{\nu} = 2976$, 2937, 2877, 1681, 1597, 1493, 1456, 1377, 1337, 1203, 1185, 1159, 1093, 1057, 1004 cm^{-1}; MS (EI): *m/z*: 387 $[M]^+$ (8.01), 330 $[M-C_2H_5CO]^+$ (7.23), 232 $[M-Ts]^+$ (100); elemental analysis calcd (%) for $C_{21}H_{25}NO_4S$: C 65.09, H 6.50, N, 3.61; found: C 64.78, H 6.74, N 3.81.

Acknowledgements

Financial support from the National Basic Research Program of China (2011CB808700) and the National Natural Science Foundation of China (20732005) is greatly appreciated. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. We thank Xiaojing Ma in this group for reproducing the results presented in entries 3 and 11 of Table 3 and Scheme 4.

Keywords: allenes · cyclization · furans · palladium · rhodium

[1] a) K. Hiroi, H. Sato, *Synthesis* **1987**, 811; b) M. E. Price, N. E. Schore, *J. Org. Chem.* **1989**, *54*, 2777; c) G. A. Kraus, Z. Wan, *Synlett* **1997**, 1259; d) Y. Siqot, P. Frère, T. Nozdryn, J. Cousseau, M. Sallé,

 © 2011 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

15213765, 2011, 18, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201003546 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

M. Jubault, J. Orduna, J. Garín, A. Gorgues, *Tetrahedron Lett.* **1997**, *38*, 1919.
[2] A. Padwa, F. R. Kinder, *J. Org. Chem.* **1993**, *58*, 21.
[3] G. Zhang, X. Huang, G. Li, L. Zhang, *J. Am. Chem. Soc.* **2008**, *130*, 1814.
[4] a) Y. Zhang, Z. Chen, Y. Xiao, J. Zhang, *Chem. Eur. J.* **2009**, *15*, 5208; b) Y. Xiao, J. Zhang, *Adv. Synth. Catal.* **2009**, *351*, 617; c) F. Liu, Y. Yu, J. Zhang, *Angew. Chem.* **2009**, *121*, 5613; *Angew. Chem. Int. Ed.* **2009**, *48*, 5505; d) Y. Bai, J. Fang, J. Ren, Z. Wang, *Chem. Eur. J.* **2009**, *15*, 8975; e) H. Gao, X. Zhao, Y. Yu, J. Zhang, *Chem. Eur. J.* **2010**, *16*, 456.
[5] D. S. Thomson, K. Koch, C. K. Hwang, S. E. Russo-Rodriguez, C. Hummel, Patent, WO 99/06410, **1999**.
[6] W. E. Bondinell, R. M. Demarinis, T. W. Ku, F. R. Pfeiffer, D. H. Shah, J. W. Venslavsky, Patent, WO 94/19354, **1994**.
[7] Y. Deng, Y. Shi, S. Ma, *Org. Lett.* **2009**, *11*, 1205.
[8] a) J. A. Marshall, E. D. Robinson, *J. Org. Chem.* **1990**, *55*, 3450; b) J. A. Marshall, G. S. Bartley, *J. Org. Chem.* **1994**, *59*, 7169; c) J. A. Marshall, E. M. Wallace, *J. Org. Chem.* **1995**, *60*, 796; d) A. S. K. Hashmi, L. Schwarz, J.-H. Choi, T. M. Frost, *Angew. Chem.* **2000**, *112*, 2382; *Angew. Chem. Int. Ed.* **2000**, *39*, 2285; e) J. T. Kim, A. V. Kel′in, V. Gevorgyan, *Angew. Chem.* **2003**, *115*, 102; *Angew. Chem. Int. Ed.* **2003**, *42*, 98; f) A. W. Sromek, A. V. Kel'in, V. Gevorgyan, *Angew. Chem.* **2004**, *116*, 2330; *Angew. Chem. Int. Ed.* **2004**, *43*, 2280; g) A. W. Sromek, M. Rubina, V. Gevorgyan, *J. Am. Chem. Soc.* **2005**, *127*, 10500; h) A. S. Dudnik, V. Gevorgyan, *Angew. Chem.* **2007**, *119*, 5287; *Angew. Chem. Int. Ed.* **2007**, *46*, 5195; i) T. Schwier, A. W. Sromek, D. M. L. Yap, D. Chernyak, V. Gevorgyan, *J. Am. Chem. Soc.* **2007**, *129*, 9868; j) S. Ma, L. Li, *Org. Lett.* **2000**, *2*, 941; k) S. Ma, Z. Yu, *Angew. Chem.* **2002**, *114*, 1853; *Angew. Chem. Int. Ed.* **2002**, *41*, 1775; l) S. Ma, J. Zhang, L. Lu, *Chem. Eur. J.* **2003**, *9*, 2447.
[9] For a pioneering study on the cyclization involving two molecules of same propadienyl ketones forming cycloisomerization and dimerization products, see: a) A. S. K. Hashmi, *Angew. Chem.* **1995**, *107*, 1749; *Angew. Chem. Int. Ed. Engl.* **1995**, *34*, 1581; b) A. S. K. Hashmi, T. L. Ruppert, T. Knöfel, J. W. Bats, *J. Org. Chem.* **1997**, *62*, 7295.
[10] For a pioneering intramolecular cyclization of symmetric substrates containing two propadienyl ketone units affording a mixture of the double-cycloisomerization products and three isomeric cyclization-conjugate addition products, see: A. S. K. Hashmi, L. Schwarz, M. Bolte, *Eur. J. Org. Chem.* **2004**, 1923.
[11] Crystal data for **2a**: $C_{21}H_{25}NO_4S$; $MW=387.48$; triclinic; space group $P\bar{1}$; final R indices ($I>2\sigma(I)$), $R1=0.0476$, $wR2=0.1306$; R indices (all data) $R1=0.0541$, $wR2=0.1390$; $a=8.9508(3)$, $b=11.2177(3)$, $c=11.2574(3)$ Å; $\alpha=91.1570(10)$, $\beta=111.7560(10)$, $\gamma=103.8230(10)°$; $V=1011.97(5)$ Å^3; $T=296(2)$ K; $Z=2$; 16168 reflections collected; 3565 unique reflections ($R_{int}=0.0199$); 3099 observations ($>2\sigma(I)$); 280 parameters. CCDC-782138 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.
[12] Crystal data for **2o**: $C_{20}H_{21}NO_4$; $MW=339.38$; monoclinic; space group $P2_1/c$; final R indices ($I>2\sigma(I)$), $R1=0.0582$, $wR2=0.1692$; R indices (all data) $R1=0.0715$, $wR2=0.1851$; $a=13.2680(3)$, $b=10.7674(3)$, $c=13.1049(4)$ Å; $\alpha=90.00$, $\beta=104.125(3)$, $\gamma=90.00°$; $V=1815.59(9)$ Å^3; $T=293(2)$ K; $Z=4$,; 11452 reflections collected; 3108 unique reflections ($R_{int}=0.0301$); 2462 observations ($>2\sigma(I)$); 235 parameters. CCDC-800852 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.
[13] a) J. Ruiz, V. Rodríguez, N. Cutillas, M. Pardo, J. Pérez, G. López, *Organometallics* **2001**, *20*, 1973; b) G. Tian, P. D. Boyle, B. M. Novak, *Organometallics* **2002**, *21*, 1462.
[14] G. A. Slough, R. G. Bergman, C. H. Heathcock, *J. Am. Chem. Soc.* **1989**, *111*, 938.
[15] These diketones were prepared by oxidation of the diols shown in Scheme 1, see: the Supporting Information of Reference [7].

Received: December 8, 2010
Published online: March 23, 2011

 © 2011 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

15213765, 2011, 18, Downloaded from https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/chem.201003546 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License